

SynchroDexterity: Rapid Non-Dominant Hand Skill Acquisition with Synchronized Guidance in Mixed Reality

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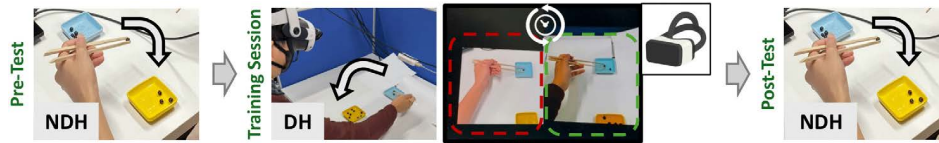


Figure 1: We introduce a novel video-see-through mixed reality system designed to mirror first-person views in real time, facilitating rapid skill acquisition for the non-dominant hand. By leveraging users' innate expertise with their dominant hand, the system tracks user's movements and overlays a synchronized video stream as real-time guidance.

ABSTRACT

Acquiring fine motor skills for the non-dominant hand is challenging yet crucial, particularly when the dominant hand is injured. While traditional mirror therapy has proven effective, it often requires extended practice and can impose significant cognitive demands. This paper presents a novel augmented reality system that leverages inverted visual feedback from the dominant hand's real-time movements, coupled with a synchronized instructional video, to facilitate efficient training for the non-dominant hand. Experimental results demonstrate that this approach significantly enhances non-dominant hand skills in a short period while maintaining a low cognitive load.

Index Terms: Fine motor skill, mixed reality, non-dominant hand, skill acquisition, visual feedback.

1 INTRODUCTION

Developing fine motor skills with the non-dominant hand (NDH) is essential when the dominant hand (DH) is injured or when tasks require coordinated bilateral use. However, the NDH often lacks the dexterity needed for precise movements, making skill acquisition challenging. Traditional training methods, such as professional coaching with specialized equipment, have demonstrated effectiveness but typically demand extended practice periods, limiting their suitability for rapid skill acquisition [1]. Visual feedback techniques, particularly mirror therapy, offer a faster alternative by reflecting DH movements to create the illusion that the NDH is performing the task, thereby facilitating motor learning [2]. However, physical mirrors are fixed in position, and interpreting inverted movements can be cognitively demanding [2, 3]. Emerging Mixed Reality (XR) technologies promise greater flexibility and adaptability by replacing mirrors with virtual simulations [4], though many existing systems support only basic movements or rely on pre-recorded expert performances that may not dynamically adapt to user needs [4, 5].

We present a novel video-see-through XR-based training

system designed to accelerate NDH skill acquisition by utilizing the user's DH as an adaptive "expert." Our system integrates inverted visual feedback from real-time DH movements with a synchronized instructional video that dynamically responds to user actions, creating a responsive, user-centered training experience aimed at reducing cognitive load and enhancing fine motor skills. We conducted a user study based on three NDH training systems: mirror-based, XR with inverted visuals, and inverted XR with synchronized expert videos.

2 METHODOLOGY

We recruited 30 right-handed participants (23 male, 7 female; mean age = 23.13, SD = 2.01) with prior experience using HMDs. Each participant was randomly assigned to one of the three groups (*Mirror*, *XR1*, *XR2*; 10 participants per group) and provided informed consent before the study.

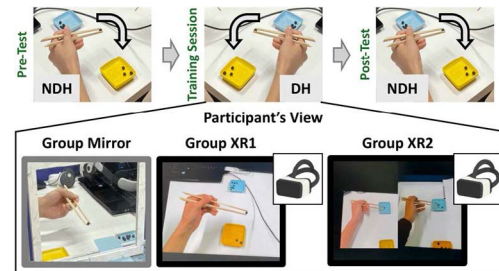


Figure 2: Experimental design.

2.1 System Setup

Participants trained under one of three conditions: *Mirror*, *XR1*, or *XR2* (Fig. 2). In *Mirror*, a physical mirror provided a horizontally inverted reflection of hand movements, mimicking standard mirror therapy [2]. In *XR1*, a full HD external webcam mounted on a Meta Quest 3 head-mounted display (HMD) provided real-time video feedback, flipped horizontally in Unity to replicate the mirror effect. An external webcam was used due to restrictions on accessing the built-in HMD camera. *XR2* is built on *XR1* by adding synchronized expert video playback aligned with user hand movements tracked by LeapMotion. In Fig. 1, the red dashed line shows the expert video and the green dashed line shows the inverted view. Implementation details for synchronization are provided below.

1. **Initialize:** $\mathbf{H}_{\text{virtual}}[0]$ (initial virtual hand position), $\mathbf{H}_{\text{start}}$ (starting boundary of the hand movement), $\mathbf{H}_{\text{end}}$ (ending

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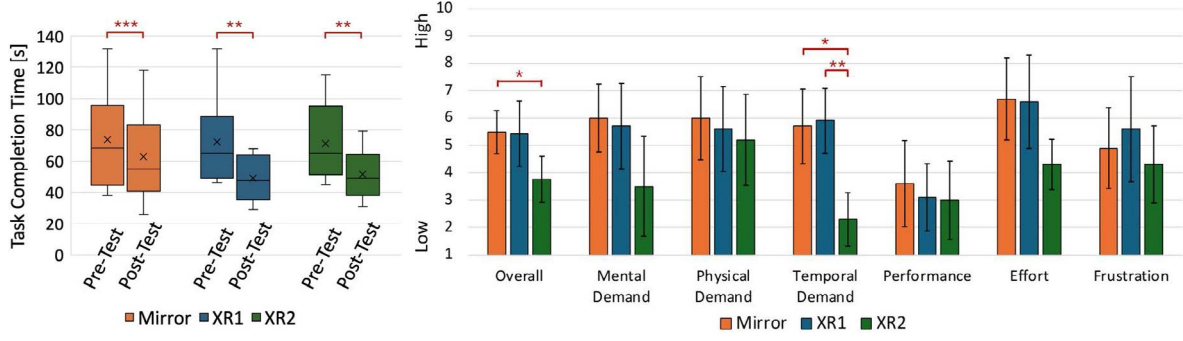


Figure 3: Effects of training systems on task performance (left) and cognitive load (right). Asterisks (*, **, and ***) denote significance levels of 0.05, 0.01, and 0.001, respectively. The bars in Fig. 3 (right) indicate 95% confidence intervals.

boundary of the hand movement), and N_{total} (total number of frames in the example video).

2. For each iteration t :

- Update the virtual hand position from the actual hand position $\mathbf{H}_{\text{real}}[t]$ retrieved from LeapMotion.
$$\mathbf{H}_{\text{virtual}}[t] = (\mathbf{H}_{\text{virtual}}[t-1] \cdot (t-1) + \mathbf{H}_{\text{real}}[t]) / t.$$
- Normalize the virtual hand position.
$$\mathbf{H}_{\text{normalized}} = (\mathbf{H}_{\text{virtual}}[t] - \mathbf{H}_{\text{start}}) / (\mathbf{H}_{\text{end}} - \mathbf{H}_{\text{start}}).$$
- Compute the video frame number.
$$n_{\text{frame}} = \lfloor \mathbf{H}_{\text{normalized}} \cdot N_{\text{total}} \rfloor.$$
- Update the video player to display frame n_{frame} .

2.2 Task and Procedure

Participants performed a chopstick-based bean-transfer task [6], using their NDH to move 10 beans from a container 20 cm away to a closer one. A bean dropped outside the target container was returned to the original container. This task, requiring precise finger coordination and pressure control, is well-suited for testing fine motor skills and aligns with our goal of providing a framework for rapid NDH training.

Task completion time was measured in each session. Training consisted of five DH trials under one of the three feedback conditions (Fig. 2), with one-minute breaks between trials. At the end, participants completed a NASA RTLX questionnaire to assess cognitive load and provided feedback on their experience via open-ended questions. Training effectiveness was evaluated using paired t -tests for pre- and post-test comparisons within groups. Between-group performance differences were analyzed via Welch's ANOVA with Games-Howell post hoc tests. Cognitive load scores were compared using one-way ANOVA or the Kruskal-Wallis test with Dunn-Sidak correction for non-parametric data.

3 RESULTS

All three groups improved significantly in NDH performance from pre- to post-test (Fig. 3 left): *Mirror* ($t(9) = 5.37$, $p < 0.001$), *XR1* ($t(9) = 3.96$, $p = 0.003$), and *XR2* ($t(9) = 3.33$, $p = 0.009$). Although Welch's ANOVA showed no significant difference in the absolute improvement among the three conditions ($F(2, 14.4) = 2.46$, $p = 0.120$), the *XR1* and *XR2* groups had larger mean improvements than the *Mirror* group and exhibited higher variability.

The cognitive load for the *XR2* system was significantly lower, particularly in temporal demand ($\chi^2 = 12.78$, $df = 2$, $p = 0.002$) and overall cognitive load ($\chi^2 = 6.19$, $df = 2$, $p = 0.045$). Although differences in other categories were not statistically significant, the *XR2* system consistently showed lower loads than the *Mirror* and *XR1* systems (Fig. 3 right).

4 DISCUSSION

All training methods improved NDH skills, but *XR2* matched mirror-based performance while reducing temporal demand, potentially easing task pace. Post-interviews revealed that participants valued *XR2*'s real-time visual assistance for confirming directional alignment, chopstick handling, and hand angles, especially during uncertainty. The synchronized video playback offered clear, step-by-step guidance that enhanced task understanding and reduced cognitive strain.

Participants also praised the XR systems for their flexible viewing angles, providing an overhead perspective and freedom of head and hand positioning. In contrast, the mirror system restricted participants to a side view of their inverted hand, differing from the direct view used in post-training performance. Overall, the XR system contributed to improved task stability and a more intuitive training experience.

5 CONCLUSION

This study demonstrates the effectiveness of our XR system for NDH skill acquisition. By leveraging users' DH expertise and providing synchronized real-time guidance, the system improved training outcomes while reducing cognitive load, making it a promising solution for fine motor skill training in diverse applications.

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